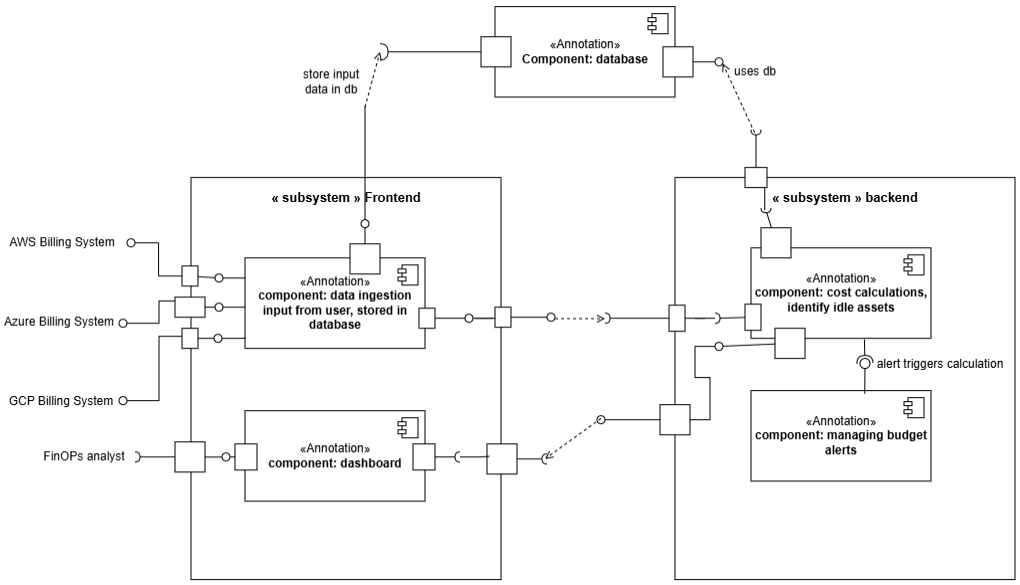
SE LAB 3

NAME: SHREYA PATIL
SRN: PES2UG24CS484
SECTION: 5H
TOPIC: MULTI CLOUD ASSET COST OPTIMIZATION PORTAL

Component diagram



Architecture Selection

Architectural Choice: Layered architecture (3 tiers)

Two Reasons:

1. Separation of responsibilities: The system separates the user side functions such as data ingestion and dashboard display from backend functions such as cost calculation, idle asset identification, and budget alert management. This makes each layer easier to develop, maintain, and modify independently.
2. Multi cloud integration: The backend provides a common processing layer between different cloud billing sources such as AWS, Azure, and GCP and the frontend. This allows billing data from different providers to be processed consistently without exposing cloud specific processing logic to the user interface.

Security Advantage:

The layered architecture provides separation and controlled access between each layers. Users

interact with the frontend rather than directly accessing the database, while the backend can make use of authentication, authorization, input validation, and access control rules before accessing stored billing and budget information.

Performance Benefit:

The separation of frontend and backend processing allows computationally intensive operations such as cost calculation and idle-asset detection to be handled by the backend, while the frontend focuses on presentation and user interaction. This reduces unnecessary processing on the client side and allows backend resources to be optimized for data processing.